

Customer Churn Prediction Using Machine Learning

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Executive Summary

This project leverages exploratory data analysis (EDA) and predictive machine learning architectures to systematically identify telecom customers at risk of churning. By deploying evaluation-ready models on a dataset of **~7,000 records**, this solution pinpoints critical churn drivers—such as contract structures and payment preferences—and delivers high-impact, actionable retention strategies to maximize customer lifetime value.

 **Key Result:** The **Logistic Regression** model delivered the top overall predictive performance, achieving an **accuracy of 80.38%** and an **ROC-AUC of 0.836**.

1. Project Objectives

- **Understand Behavior:** Conduct deep-dive Exploratory Data Analysis (EDA) to map customer trends and systemic behaviors.
- **Isolate Drivers:** Identify the exact demographic, subscription, and billing factors influencing churn.
- **Predictive Modeling:** Build and optimize robust machine learning models to forecast customer defection.
- **Performance Benchmarking:** Statistically compare model performance metrics to ensure deployment reliability.
- **Strategic Retention:** Formulate targeted business recommendations to directly improve retention.

2. Dataset & Core Stack

Dataset Description: Telco Customer Churn Dataset containing approximately **7,000 customer records** spanning demographic metrics, subscription components, billing structures, and historical churn indicators.

Python

Pandas

NumPy

Matplotlib

Seaborn

Scikit-learn

Google Colab

3. Methodology & Execution Pipeline

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Data Preparation Removed missing values, standardized types, and dropped identifiers (customerID).	Exploratory Analysis Profiled churn behavior against demographics, charges, and contract formats.	Feature Eng. Transformed categorical variables into numerical structures via One-Hot Encoding.	Modeling & Eval Trained candidate models and evaluated using standard classification metrics.
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4. Key Exploratory Data Analysis Insights

- **Baseline Churn Rate:** **26.58%** of the total customer base actively churned.
- **Contract Frameworks:** Customers on **Month-to-month contracts** exhibited the highest propensity to churn.
- **Infrastructure Discrepancies:** **Fiber optic** subscribers churned significantly more frequently than DSL users.
- **Billing Friction:** Users relying on **Electronic checks** demonstrated disproportionately higher churn rates.
- **Financial Thresholds:** High monthly charges strongly correlated with an increased probability of churn.
- **Loyalty Stabilizer:** **Long-tenure customers** showed exceptional retention and vastly lower churn metrics.

5. Model Performance Benchmarking

MACHINE LEARNING MODEL	ACCURACY	PRECISION	RECALL	F1-SCORE	ROC-AUC
Logistic Regression 🏆	80.38%	64.76%	57.49%	60.91%	0.836
Random Forest	79.03%	63.30%	50.27%	56.04%	0.817

Primary Predictors (Ranked by Feature Importance Scale)

- 1. Total Charges
- 2. Tenure
- 3. Monthly Charges
- 4. Electronic Check
- 5. Fiber Optic Internet
- 6. Contract Type
- 7. Online Security
- 8. Paperless Billing

6. Strategic Business Recommendations

- **Contractual Migration:** Incentivize month-to-month users to adopt long-term agreements through targeted loyalty discounts.
- **Service Remediation:** Audit and improve the customer onboarding/quality experience specifically for Fiber Optic subscribers to decrease attrition.
- **Frictionless Billing:** Run campaigns encouraging a transition from manual electronic checks to automated payment methods.
- **Early-Stage Interventions:** Deploy high-touch customer success and proactive retention strategies during the critical **first 12–18 months** of tenure.
- **Value Tier Safeguards:** Introduce proactive loyalty rewards and dedicated account perks for high-value/high-charge tiers.
- **Security Bundling:** De-risk churn by bundling technical support and online security directly into core subscriptions.

7. Conclusion

The **Logistic Regression** framework establishes a highly reliable predictive baseline for this dataset, leading with **80.38% accuracy** and an **ROC-AUC of 0.836**. Implementing this model in production allows the business to proactively flag high-risk customers, shifting customer management from a reactive posture to a targeted, data-driven retention model.